

Figure 1 | Spatial geometry in weather modelling

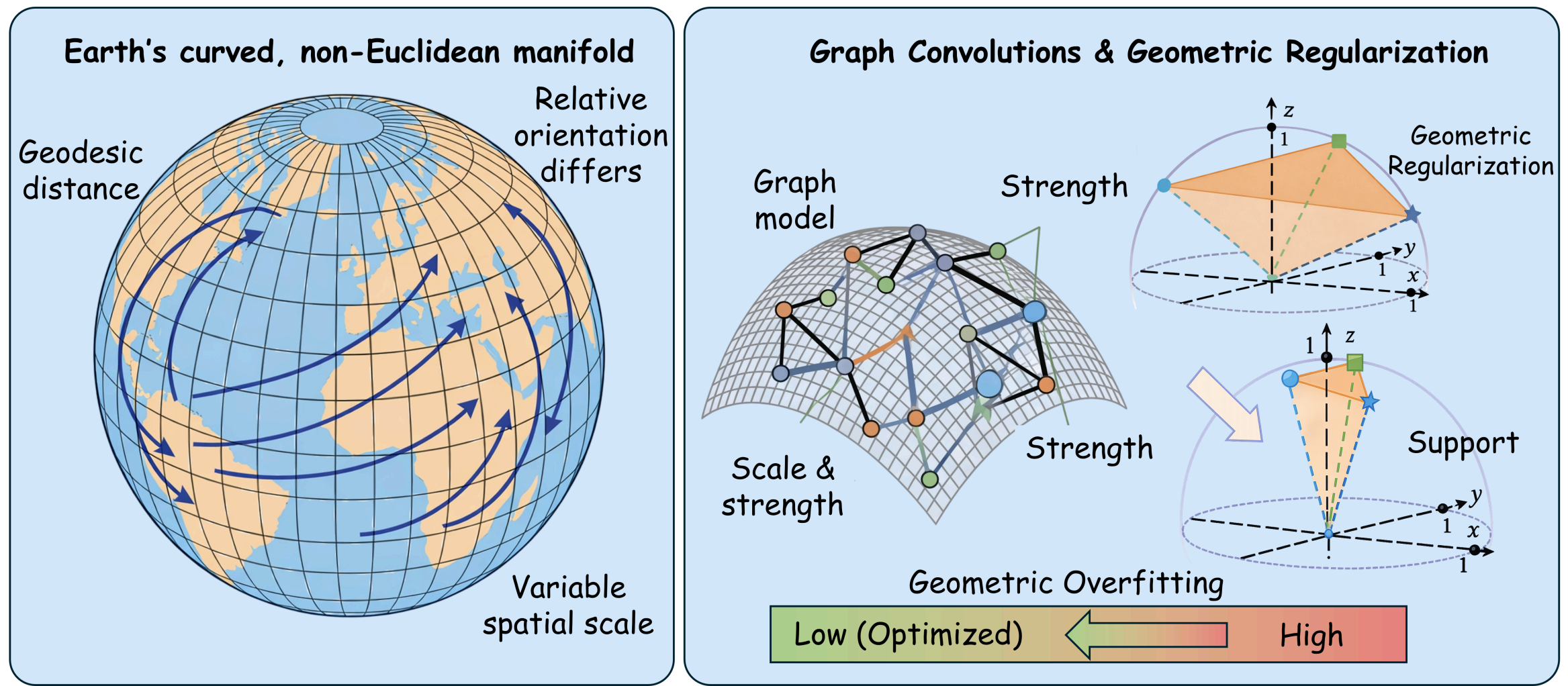


Figure 1. Conceptual illustration of spatial geometry in weather modelling. The left panel illustrates variation in geodesic distance and relative orientation on the sphere; the right panel schematically depicts graph convolution and geometry-sensitive weighting. All surfaces, node positions and colour gradients are illustrative and are not measured forecasts or quantitative evidence of improved regularization.

Figure 2 | Implementation-aligned model architecture

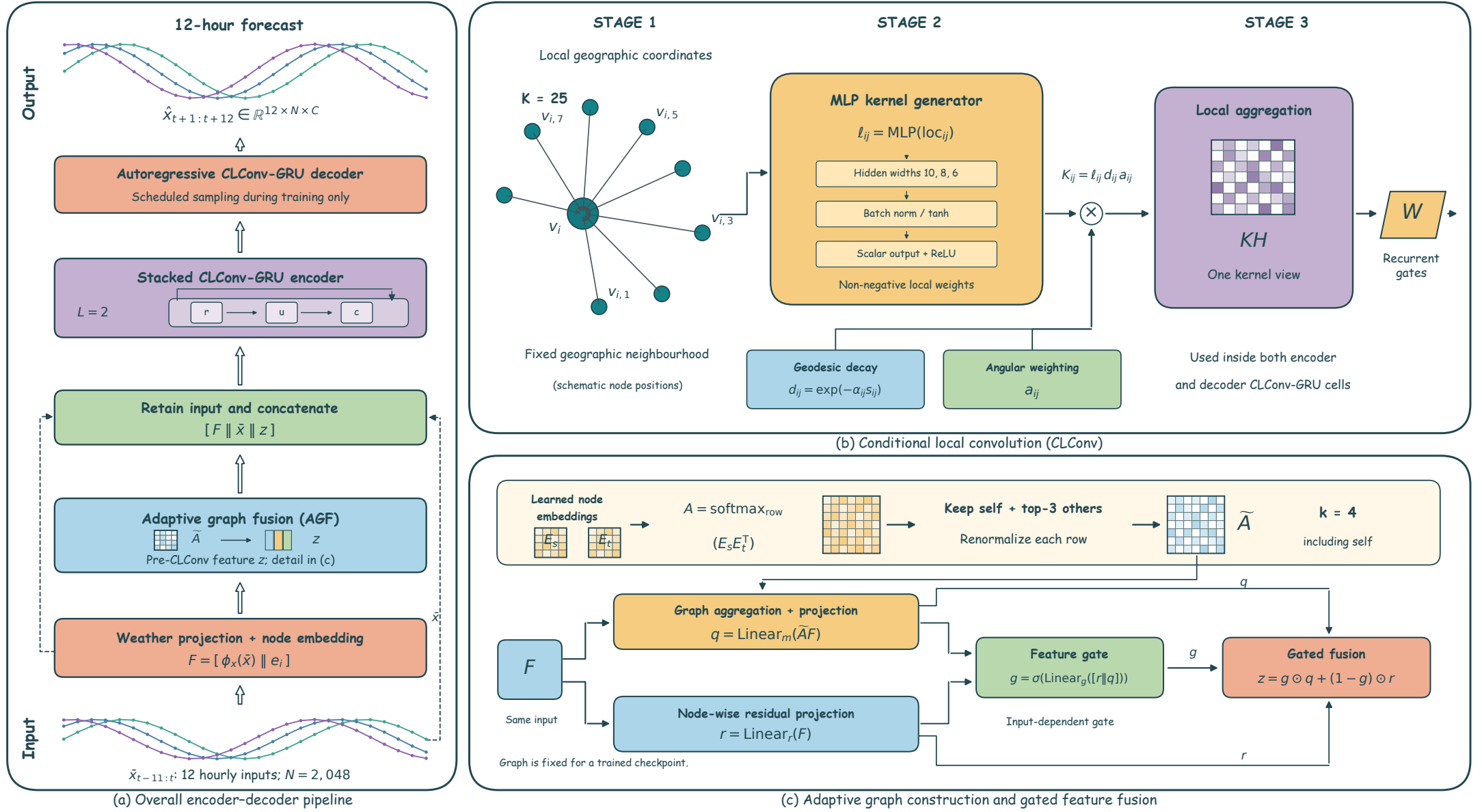


Figure 2. Implementation-aligned CLCRN-AGF architecture. (a) Standardized weather inputs are projected and concatenated with learned node embeddings to form F . AGF acts on F before the recurrent backbone; its output z is concatenated with F and the standardized weather input. A two-layer CLConv-GRU encoder and autoregressive decoder predict the next 12 hours. Scheduled sampling is used only during training. (b) The local kernel combines learned coordinate-dependent weights, geodesic decay and angular weighting over 25 geographic neighbours. These experiments use a single kernel view in the recurrent cells. (c) Row-wise softmax of learned source and destination embeddings defines an input-independent adjacency for each checkpoint. Self plus the three strongest other neighbours are retained and renormalized. Graph-aggregated and residual projections q and r use the same pre-CLConv F ; their input-dependent gate forms $z = g * q + (1 - g) * r$. The control omits z and retains the original weather and node embeddings. Traces, node positions and matrix cells are illustrative.

Figure 3 | Current-split forecasting comparison

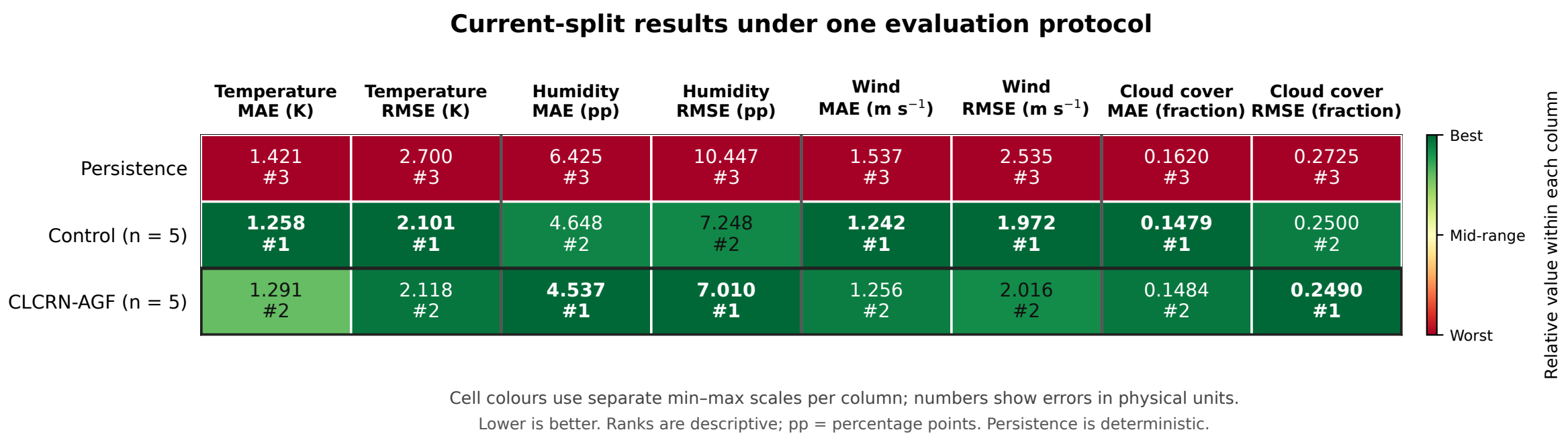


Figure 3. Current-split cumulative 12-hour MAE and RMSE for persistence and the two trained variants. Control and CLCRN-AGF values are means across five training seeds; persistence is deterministic. All results use the same zero-inclusive, spatially unweighted evaluator and agree with the current context table. Colours use independent min-max normalization within each task/metric column; the numbers retain physical units. Rankings are descriptive, not significance tests. Temperature is in K, relative humidity in percentage points (pp), wind in m s⁻¹, and cloud cover in fraction.

Figure 4 | Five-seed primary comparison

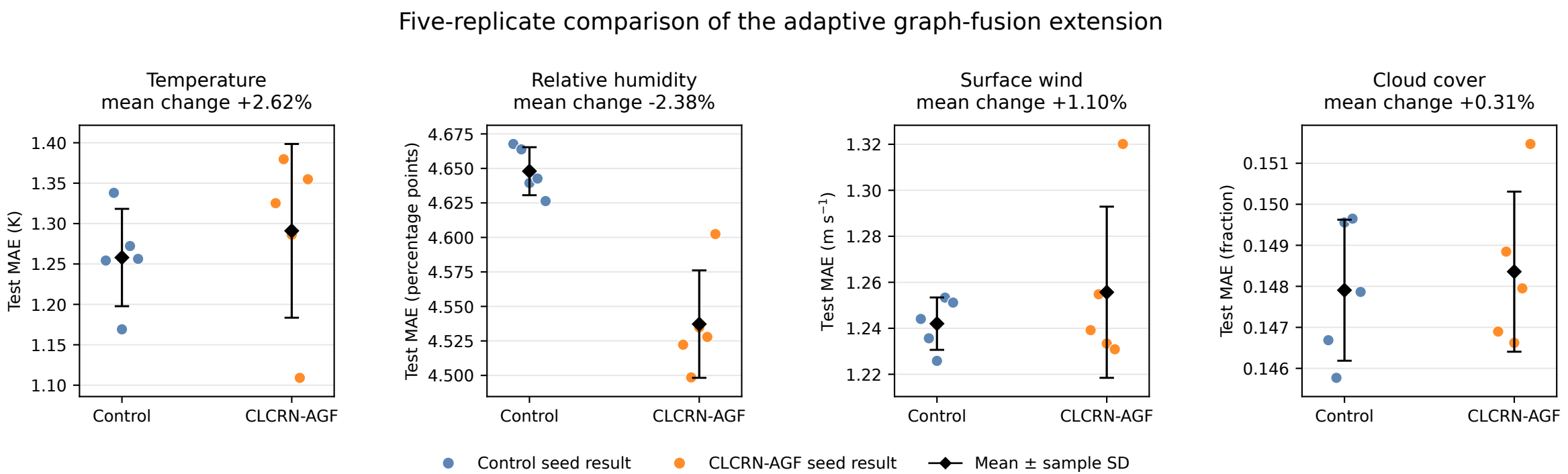


Figure 4. Cumulative 12-hour test MAE across five independently trained replicates for each variant and task. Coloured points show individual seeds; black diamonds and bars show the mean plus or minus sample standard deviation (ddof = 1). The percentage above each panel is $100 \times (\text{AGF mean} / \text{control mean} - 1)$; a negative value denotes lower MAE. These are the same 40 primary checkpoints used in the current statistical table. Seed labels are not treated as matched random streams. Panels have task-specific units and focused vertical scales.

Figure 5 | Forecast errors across horizons

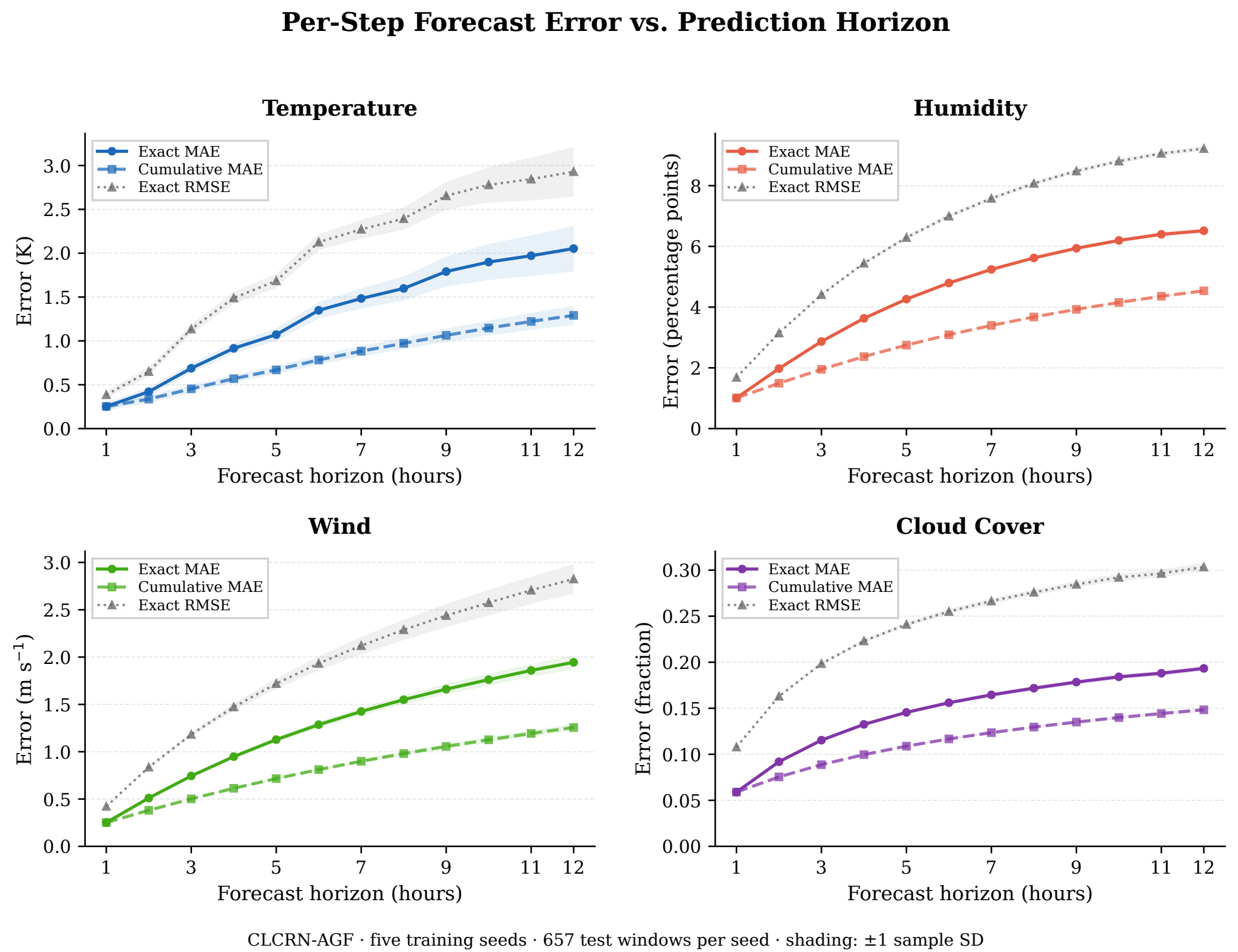


Figure 5. Per-step forecast error of five CLCRN-AGF models per task on all 657 test windows. Solid circles show exact-lead MAE, dashed squares show MAE accumulated over leads 1 through h , and grey triangles show exact-lead RMSE. Lines denote the training-seed mean and shading denotes one sample standard deviation ($\text{ddof}=1$). These 20 AGF checkpoints are the AGF half of the 40 primary runs in Table 1. At 12 h the cumulative MAE equals the corresponding Table 1 value. Independent vertical scales retain each task's physical units. Valid zero targets are included and spatial weights are uniform.

Figure 6 | Comparison under one training protocol

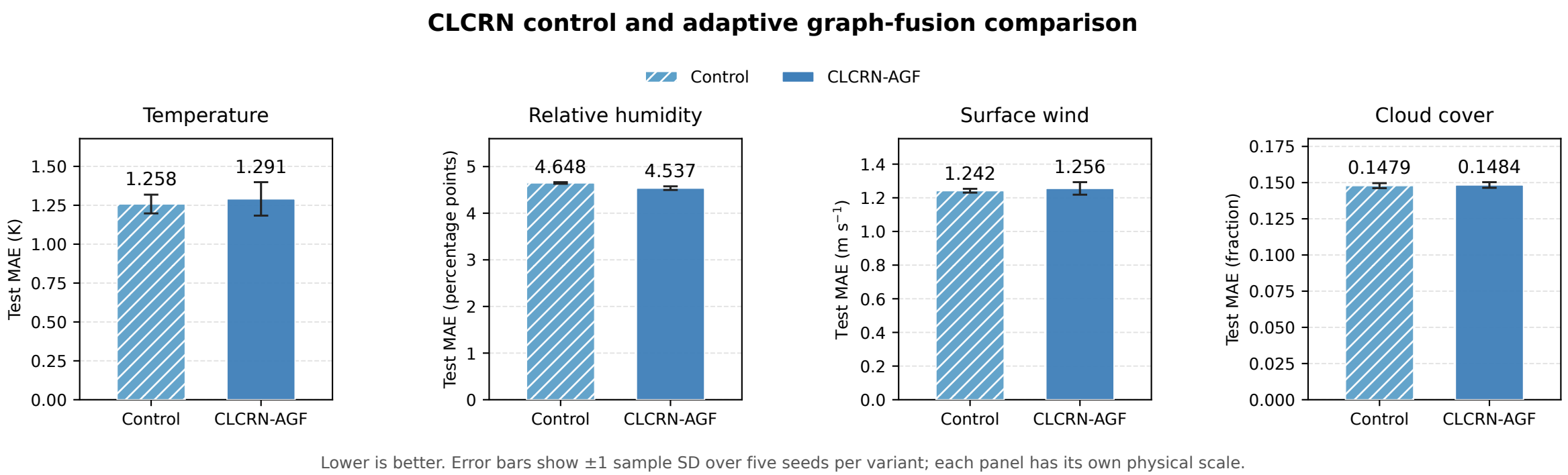


Figure 6. Cumulative 12-hour test MAE for CLCRN control and CLCRN-AGF under the same current training and evaluation protocol. Bars show five-seed means; error bars show plus or minus one sample standard deviation (ddof = 1). These values exactly match the primary statistical table and Figure 4. All axes start at zero; panels use different physical units and independent scales. This comparison does not isolate a training-schedule effect.

Figure 7 | Sensitivity to missing input nodes

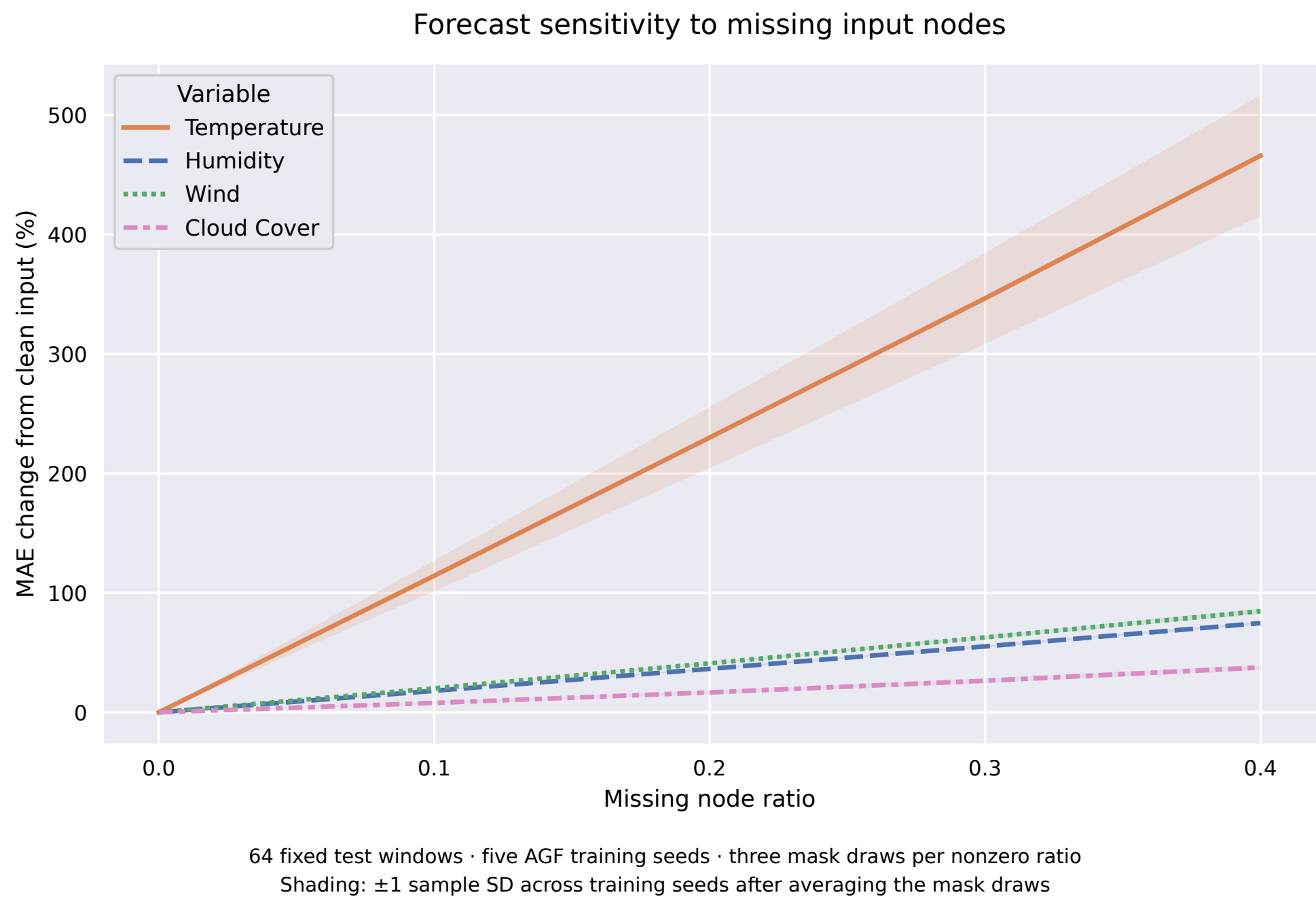
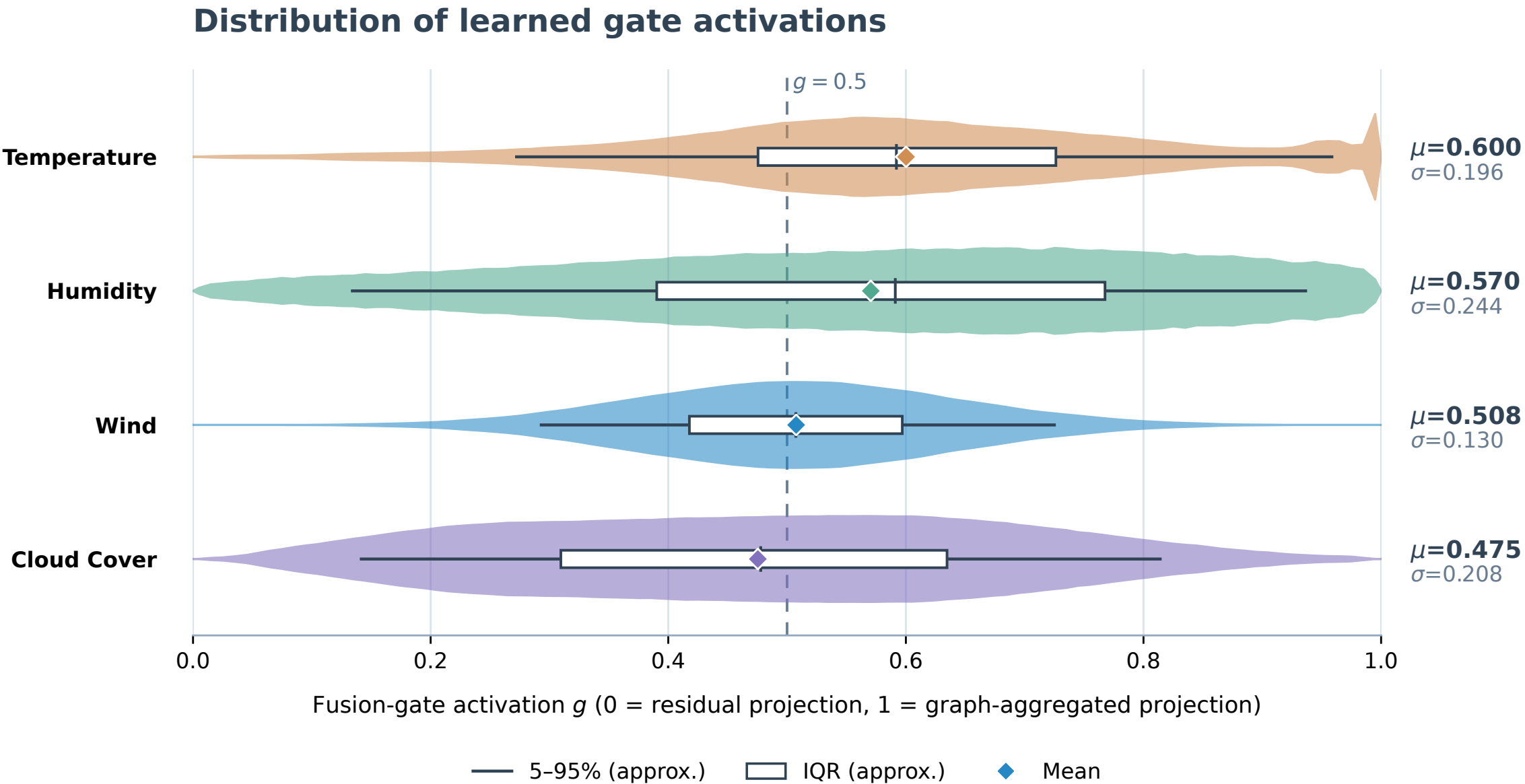


Figure 7. Missing-input-node sensitivity of five AGF checkpoints per task, evaluated on the same fixed subset of 64 test windows at every ratio. The same node mask is applied across all 12 input hours and signal channels; standardized values are set to zero, corresponding to training-mean imputation. The plotted quantity is 100 times (masked MAE / clean MAE - 1), with each checkpoint's clean MAE measured on that same subset. At each nonzero missing-node ratio, mask seeds 11, 22 and 33 are averaged within a training seed before computing the mean and sample standard deviation across five training seeds. The clean point is zero by definition. Shaded bands indicate descriptive training-seed variation, not confidence intervals or a significance test. Tasks use distinct colours and line patterns. This diagnostic uses 64 of the 657 test windows.

Figure 8 | Distribution of learned gate activations



Five AGF seeds $\times$ 657 test windows $\times$ 12 input steps $\times$ 2,048 nodes $\times$ all 8 gate channels

Figure 8. Distribution of scalar gate activations for all four tasks. Each task pools 645,857,280 activations from five AGF checkpoints, all 657 test windows per checkpoint, 12 input steps, 2,048 nodes and all eight channels, without channel averaging. Mirrored profiles interpolate the centres of the recorded 100-bin histogram on [0,1]; widths are normalized separately within each task. White boxes, centre lines and whiskers are approximate 25th-75th, 50th and 5th-95th percentiles computed by linear interpolation of the binned cumulative counts. Diamonds and right-side mean and population standard deviation are the exact full-stream statistics reported in Table 4. The dashed line is $g=0.5$. The gate combines a graph-aggregated projection and a residual linear projection of the same pre-CLConv input.

Table 1. Five-replicate comparison of CLCRN-AGF and the schedule-matched CLCRN control. Values are mean $\pm$ sample standard deviation across seeds 2021–2025. Differences are CLCRN-AGF minus control; negative values favour CLCRN-AGF. Confidence intervals and two-sided p -values use Welch’s unequal-variance approximation.

Task	Metric	Control (mean $\pm$ s)	CLCRN-AGF (mean $\pm$ s)	Difference [95% CI]	p_{Welch}
Temperature	MAE	1.258 $\pm$ 0.060	1.291 $\pm$ 0.107	+0.033 [−0.100, +0.166]	0.5701
	RMSE	2.101 $\pm$ 0.045	2.118 $\pm$ 0.129	+0.017 [−0.141, +0.174]	0.7942
Relative humidity	MAE	4.648 $\pm$ 0.017	4.537 $\pm$ 0.039	−0.111 [−0.158, −0.063]	0.0015
	RMSE	7.248 $\pm$ 0.043	7.010 $\pm$ 0.055	−0.239 [−0.311, −0.166]	8.17×10^{-5}
Surface wind	MAE	1.242 $\pm$ 0.011	1.256 $\pm$ 0.037	+0.014 [−0.032, +0.059]	0.4699
	RMSE	1.972 $\pm$ 0.023	2.016 $\pm$ 0.093	+0.043 [−0.071, +0.158]	0.3657
Cloud cover	MAE	0.1479 $\pm$ 0.0017	0.1484 $\pm$ 0.0020	+0.0005 [−0.0022, +0.0031]	0.7077
	RMSE	0.2500 $\pm$ 0.0017	0.2490 $\pm$ 0.0028	−0.0010 [−0.0045, +0.0025]	0.5103

All scores include valid zero targets and use equal spatial weights over the full 12-hour forecast and 657 test examples. Errors are in K for temperature, percentage points for relative humidity, m s^{-1} for surface wind, and fractions for cloud cover. Benjamini–Hochberg correction across the eight comparisons retains only the two humidity differences at $q < 0.05$ (MAE $q = 0.00604$; RMSE $q = 0.000653$).

Table 2. Contextual test results on the same test split. Persistence is deterministic; control and CLCRN-AGF are means across five seeds. All methods use the same zero-inclusive, spatially unweighted evaluation over the full 12-hour forecast. Lower is better.

Method	Temperature		Relative humidity		Surface wind		Cloud cover	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Persistence	1.421	2.700	6.425	10.447	1.537	2.535	0.1620	0.2725
CLCRN control (5 seeds)	1.258	2.101	4.648	7.248	1.242	1.972	0.1479	0.2500
CLCRN-AGF (5 seeds)	1.291	2.118	4.537	7.010	1.256	2.016	0.1484	0.2490

Errors are in K for temperature, percentage points for relative humidity, m s^{-1} for surface wind, and fractions for cloud cover. Variation across model seeds is reported in Table 1.

Table 3. Component ablations with a matched 100-epoch FP32 training protocol. All entries report mean $\pm$ sample standard deviation over seeds 2023–2025 on the full 12-hour test forecast. Lower is better.

Budget	Task	Variant	MAE	RMSE
100 epochs, 3 seeds	Temperature	Control (no AGF)	1.233 ± 0.056	2.113 ± 0.043
		w/o gated fusion	1.267 ± 0.066	2.047 ± 0.103
		Full CLCRN-AGF	1.250 ± 0.127	2.057 ± 0.138
100 epochs, 3 seeds	Relative humidity	Control (no AGF)	4.636 ± 0.009	7.227 ± 0.043
		w/o gated fusion	4.599 ± 0.024	7.132 ± 0.062
		Full CLCRN-AGF	4.555 ± 0.041	7.040 ± 0.048

The control removes the entire AGF module, including its residual projection, changing the encoder input width and parameter count. The ungated variant uses a fixed equal average of the graph and residual branches. The control and full-model rows use the same seed subset from Table 1. Errors are in K for temperature and percentage points for relative humidity.

Table 4. Gate-activation statistics pooled over all five CLCRN-AGF seeds and the full test set. The gate balances a graph-aggregated projection and a residual linear projection of the same input, before CLConv. Standard deviations describe the pooled population of scalar gates.

Variable	Mean μ	Std σ	Min	Max
Temperature	0.600	0.196	0.001	1.000
Relative humidity	0.570	0.244	0.001	1.000
Surface wind	0.508	0.130	0.049	0.954
Cloud cover	0.475	0.208	0.002	0.997

Each task includes $5 \times 657 \times 12 \times 2,048 \times 8 = 645,857,280$ scalar gates (seeds, test examples, input steps, nodes and channels). Channels are retained individually; no channel averaging or test-set subsampling is applied.